

2025 Chapter Review Vector Operation

- Notice: 1. Make sure that you have filled the form on the left side of seal line.
2. Write your answers on the exam paper.
3. This is a close-book exam.
4. The exam with full score of 100 points lasts 120 minutes.

Question No.											Sum
Score											

(1) Find the distance between two line $l_1 : \begin{cases} x - y = 0 \\ z = 0 \end{cases}$, and $l_2 : \frac{x-2}{4} = \frac{y-1}{-2} = \frac{z-3}{-1}$;

(2) Find the distance between point and line $M(1,0,2)$ and $l_2 : \frac{x}{2} = \frac{y+1}{-2} = \frac{z-1}{1}$;

(2) $z - e^x + 2xy = 3$ find tangent plane at $(1,2,0)$;

(3) $\vec{F} = yzf(xyz)\vec{i} + xzf(xyz)\vec{j} + xyf(xyz)\vec{k}$, Find $\text{curl}(\vec{F})$;

(4) $u = x^2 + e^y z$, and $xz + \ln(1 + yz) = 1$. $\vec{d} = 3\vec{i} - 4\vec{j}$;
Find directional derivative of function $u = u(x, y)$ at $(1, 0)$

(5) If $\text{grad}f(x, y) = \langle \frac{y}{1+x^2y^2}, \frac{x}{1+x^2y^2} \rangle$ Find $f(x, y)$

(6) If $x^2 + 2y^2 + 3z^2 = 21$ Find normal line at $(1, -2, 2)$

(7) $l_1 : \begin{cases} 2x^2 + 3y^2 + z^2 = 9 \\ z^2 = 3x^2 + y^2 \end{cases}$ Find normal plane and tangent line at $(1, -1, 2)$

(8) $3x^2 + y^2 = 1$, Find surface equation revolve to x -axis

(9) $l_1 : \begin{cases} x - y + z = 0 \\ 2x - y + 3z = 0 \end{cases}$ and $S : x^2 + y^2 + z^2 = 4$

Find equation plane perpendicular to l_1 line and tangent to surface S

(10) Find the symmetric point of $M(0,0,0)$ to the line $\frac{x}{2} = \frac{y+1}{1} = \frac{z-4}{-2}$

